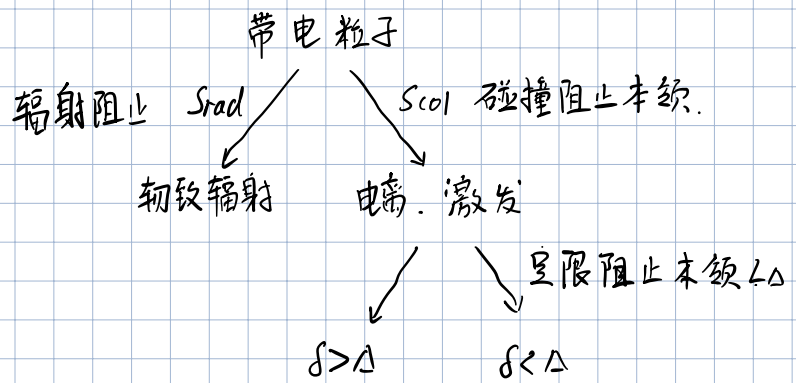


$$\frac{E^2-m_0^2c^4}{c^2} = \left(\frac{E_\gamma}{c}\right)^2 + \left(\frac{E_\gamma'}{c}\right)^2 - \frac{2E_\gamma E_\gamma'}{c^2}\cos\theta$$

$$E^2 - m_0^2c^4 = E_\gamma^2 + E_\gamma'^2 - 2E_\gamma E_\gamma' \cos\theta.$$

$$E_\gamma - E_\gamma' = E - m_0c^2$$

$$E = E_\gamma - E_\gamma' + m_0c^2$$

$$(E_\gamma - E_\gamma' + m_0c^2)^2 - m_0^2c^4 = E_\gamma^2 + E_\gamma'^2 - 2E_\gamma E_\gamma' \cos\theta.$$

$$2(E_\gamma - E_\gamma')m_0c^2 - 2E_\gamma E_\gamma' = -2E_\gamma E_\gamma' \cos\theta.$$

$$(E_\gamma - E_\gamma')m_0c^2 = (1-\cos\theta)E_\gamma E_\gamma'$$

$$\frac{1}{E_\gamma} - \frac{1}{E_\gamma'} = \frac{1}{m_0 c^2}(1-\cos\theta).$$

$$E_\gamma' = \frac{E_\gamma}{\frac{1-\cos\theta}{m_0 c^2} E_\gamma + 1} = \frac{E_\gamma m_0 c^2}{(1-\cos\theta)E_\gamma + m_0 c^2}$$

$$\frac{E_\gamma^2}{\frac{m_0 c^2}{2} + E_\gamma} \qquad \frac{E_\gamma^2}{\infty}.$$

$$3x^{-2} + 9x^3 = x^2$$

$$9x^5 = x^4 - 3$$

$$x = \left(\frac{x^4 - 3}{9}\right)^{\frac{1}{5}}$$

$$x_2 = \left(\frac{x_1^4 - 3}{9}\right)^{\frac{1}{5}}$$

$$x_{n+1} = \left(\frac{x_n^4 - 3}{9}\right)^{\frac{1}{5}}$$